

Risk-Calibrated Fallback: A Selective-Prediction Approach to Minimum-Risk-Manoeuvre Triggering in Closed-Loop Autonomous Driving

Hassan Waqas Saleem, Rahat Zeeshan Saleem, and Hina Maryam

Abstract—Every deployed autonomous vehicle has a minimum-risk manoeuvre (MRM) — pull over, slow down, hand control back to a driver — and when to trigger it is almost universally a hand-tuned threshold on a confidence score. That score is precisely the quantity known to be untrustworthy under distribution shift, so the trigger for the safety behaviour rests on a calibration that can silently expire exactly when the vehicle needs it most. Prior conformal-prediction work in this space is almost entirely open-loop: it produces prediction regions and stops, so it cannot see that a conservative region drives an MRM trigger, and that triggering itself has a safety cost — freezing in a live lane. We formalise MRM triggering as a selective-prediction problem with a guarantee on the *decision*, not merely on the *prediction*, using Learn-then-Test to certify a trigger threshold against a bounded miss-rate target under exchangeability, and Conformal Risk Control as a tighter alternative when the closed-loop loss is empirically monotone in the threshold. We evaluate in closed loop on real, replayed driving scenarios (Argoverse 2, nuScenes) under reactive traffic, using a fallback manoeuvre compliant with UN Regulation No. 157’s minimum-risk-manoeuvre deceleration bound and harm labelled by the nuPlan closed-loop safety definition (at-fault collision or time-to-collision below 0.95 s). We report both the missed-intervention rate the calibrated trigger is built to bound and the unnecessary-stop rate open-loop evaluation cannot see, and we show that the two are inseparable: any trigger design that only reports one is reporting half the picture.

Index Terms—Autonomous driving, conformal prediction, selective prediction, minimum-risk manoeuvre, closed-loop safety evaluation, trajectory prediction, risk calibration, distribution shift.

I. INTRODUCTION

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk manoeuvre: slow to a stop, pull to the shoulder, or otherwise hand control back to a human. The decision of *when* to invoke this behaviour is, in nearly every deployed and published system, a hand-tuned threshold on some scalar confidence or uncertainty score produced by a perception or prediction module. That practice has a specific and largely unexamined weakness: the very quantity the threshold

is applied to — a learned model’s uncertainty estimate — is exactly what has been shown to degrade under the kind of distribution shift a vehicle experiences when it enters a new city, a new weather condition, or simply a stretch of road unlike anything in its calibration set. A threshold tuned under one operating distribution carries no guarantee once that distribution has moved, and the failure is silent: the system continues to report a confidence number, and nothing in the number itself signals that the number is now wrong.

Conformal prediction offers a principled alternative: rather than a point confidence score, it produces a prediction region with a distribution-free, finite-sample coverage guarantee under exchangeability. A substantial and fast-growing literature has applied conformal prediction to trajectory forecasting and to planning under uncertainty in autonomous driving. Almost without exception, however, this literature is *open-loop*: it constructs a prediction region, reports its coverage and its size, and stops. Two things are invisible from that vantage point. First, a region’s effect on a downstream control decision — specifically, whether it is used to trigger a fallback — is not evaluated at all; open-loop coverage and closed-loop safety outcome are simply assumed to move together. Second, and more consequentially, *triggering the fallback is not free*. A vehicle that stops in a live lane because its uncertainty region grew large is not obviously safer than one that continues; an evaluation that reports only coverage, or only collision rate under the fallback, cannot see the trade-off between the two, and cannot see that a policy which triggers whenever it is asked to would trivially achieve a good miss rate while being useless.

This paper studies fallback triggering as a decision problem with a formal guarantee attached to the decision itself, evaluated where that guarantee actually has to hold: in closed loop, on real driving scenarios, with a trigger threshold that has consequences for what the vehicle does next. We make three contributions.

- 1) We formalise minimum-risk-manoeuvre triggering as *selective prediction* with a guarantee on the closed-loop miss rate — the probability that a harmful event occurs without the fallback having fired first — rather than on marginal coverage of a static prediction region. We calibrate the trigger threshold with Learn-then-Test [2], which requires no assumption that the closed-loop loss be monotone in the threshold, and report Conformal Risk Control [3] as a tighter alternative on the scenarios where monotonicity is observed to hold.

H. W. Saleem (corresponding author) is with the School of Electrical Engineering and Computer Science (SEECs), National University of Sciences and Technology (NUST), H-12, Islamabad 44000, Pakistan (e-mail: hsaleem.msee20seecs@seecs.edu.pk).

R. Z. Saleem is with the NUST Business School (NBS), National University of Sciences and Technology (NUST), H-12, Islamabad 44000, Pakistan (e-mail: rahat.mscm21nbs@student.nust.edu.pk).

H. Maryam is with the State Bank of Pakistan, I.I. Chundrigar Road, Karachi 74000, Pakistan (e-mail: hina.maryam@sbp.org.pk).

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- 2) We evaluate entirely in closed loop, on scenarios replayed from real driving logs (Argoverse 2 and nuScenes, via ScenarioNet [8] and MetaDrive) under reactive traffic, and we report the unnecessary-stop rate alongside the missed-intervention rate at every operating point — the over-conservatism trade-off that an open-loop study structurally cannot see.
- 3) We connect the closed-loop outcome to the underlying predictor’s calibration under dataset shift, asking directly whether a degradation in open-loop conformal coverage translates into a measurable degradation in closed-loop safety, and report the answer honestly whether or not it supports the premise.

The remainder of the paper is organised as follows. Section II situates this work against open-loop conformal prediction for driving, task-relevant failure detection, and closed-loop safety evaluation more broadly. Section III formalises the triggering problem and the calibration procedure. Section IV describes the closed-loop evaluation environment, the fallback manoeuvre, and the harm labelling used to score every rollout. Section V reports results. Section VI discusses limitations, and Section VII concludes.

II. RELATED WORK

A. Conformal prediction for autonomous driving

Conformal prediction [1] constructs prediction sets with a distribution-free, finite-sample marginal coverage guarantee under exchangeability between calibration and test data, and has been applied extensively to trajectory forecasting and to safe planning under uncertainty. Lindemann *et al.* embed conformal prediction regions directly into a model predictive controller to obtain probabilistic safety guarantees, but their guarantee collapses once the deployment distribution shifts away from the calibration distribution — a limitation Rahaman *et al.* [4] address with robust conformal prediction driven by a conditional generative model of the shift, evaluated in a synthetic multi-agent corridor. SafePath [5] integrates conformal prediction into an LLM-based path-planning pipeline, providing a guarantee that a set of candidate paths contains a safe one with a user-specified probability, and reports both open-loop (nuScenes) and closed-loop (highway-env, synthetic traffic) results. In all three cases the guarantee is on the *prediction* — a region contains the truth, or a path set contains a safe path — rather than on the eventual control decision the region is used to make, and none evaluates a fallback-triggering decision on real, replayed driving logs under reactive traffic. This paper’s guarantee is instead on the decision: the probability that a minimum-risk manoeuvre fires before a harmful event, evaluated end-to-end in closed loop.

B. Task-relevant and closed-loop failure detection

Farid *et al.* [6] propose a run-time monitor for trajectory-prediction failures that is *task-relevant* by construction: it flags a misprediction only when the resulting planning cost is anomalous relative to the predicted cost distribution, with data-free false-positive and false-negative rate bounds. Their detector is evaluated against hand-labelled scenarios in an open-loop

and semi-reactive setting, and — like the conformal-prediction literature above — does not close the loop through an actuated fallback and re-measure the outcome. Their central insight, that a task-irrelevant misprediction should not count against a detector, is reflected in this paper’s harm labelling: a collision the ego vehicle suffers while already stopped or while being struck from behind is scored as a distinct *induced-collision* cost of the fallback itself, never as a missed intervention.

C. Risk-sensitive safety analysis

Chapman *et al.* [7] develop risk-sensitive safe sets using Conditional Value-at-Risk to assess the severity, not merely the probability, of constraint violation along a stochastic trajectory, providing the risk-sensitive vocabulary this paper borrows in motivating why a mean missed-intervention rate is the appropriate quantity to control at a chosen operating point, even though a tail-severity extension (Section VI) is left to future work.

D. Closed-loop simulation infrastructure

ScenarioNet [8] unifies heterogeneous real-world driving logs — including Argoverse 2, nuScenes, nuPlan, and Lyft Level 5 — into a common scenario description that can be replayed interactively in the MetaDrive simulator, with either logged-trajectory replay or reactive rule-based (IDM) traffic. This work is built directly on that infrastructure: every scenario reported in Section V is a real logged driving scenario, replayed headless with reactive traffic, not a synthetic or hand-crafted one.

III. PROBLEM FORMULATION AND METHOD

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ : at every decision tick the vehicle computes an uncertainty score $u_t(X)$ from a trajectory predictor’s output for nearby agents, and engages the minimum-risk manoeuvre the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{I}[\text{harm in } X \text{ under } \lambda, \text{ MRM not first}], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{I}[\text{MRM fires under } \lambda, X \text{ harm-free at } \lambda = \infty], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1, \dots, X_n} \left(\mathbb{E}_X [L(X, \hat{\lambda})] \leq \alpha \right) \geq 1 - \delta, \quad (3)$$

while minimising $\mathbb{E}_X [S(X, \hat{\lambda})]$ — the smallest autonomy loss compatible with the risk target α .

Equation (3) is not a conformal coverage statement: $L(X, \lambda)$ depends on the entire closed-loop rollout under λ , since engaging the fallback changes every subsequent state, so it is neither a marginal coverage event on a static prediction

nor guaranteed to be monotone in λ a priori. We therefore calibrate $\hat{\lambda}$ with Learn-then-Test [2]: for a finite grid $\lambda_1 > \lambda_2 > \dots > \lambda_J$ ordered from least to most conservative, compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n} \sum_i L(X_i, \lambda_j)$ on n calibration scenarios, a Hoeffding–Bentkus p -value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, and apply fixed-sequence testing starting from the most conservative λ ; every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ , and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). Where the empirical loss is observed to be monotone in λ across the calibration set, we additionally report Conformal Risk Control [3], whose expectation guarantee is tighter but requires that monotonicity assumption; where it is violated we report the violation rate directly rather than silently falling back to the weaker method.

B. Trigger score

At each decision tick the trigger score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle’s own near-term extent. Writing the ego’s constant-heading, constant-speed extrapolation over a horizon of T steps as $\{p_\tau^{\text{ego}}\}_{\tau=1}^T$ — the same route-agnostic motion model used by the harm metric in Section IV-B, deliberately *not* the vehicle’s planned, route-following path, so that the trigger cannot credit itself for steering behaviour the safety label does not assume — and a predictor’s K -mode forecast for agent i as $\{\hat{p}_\tau^{(i,k)}\}$, each predicted offset is decomposed into a longitudinal component $a_\tau^{(i,k)}$ (along the ego heading) and a lateral component $\ell_\tau^{(i,k)}$ (perpendicular to it), each compared against a separate margin: a length-based longitudinal margin $m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and a width-based lateral margin $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion at (i, k, τ) is

$$\rho_\tau^{(i,k)} = \max \left(0, 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}} \right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}} \right)^2} \right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. A single isotropic clearance radius cannot represent both margins without either missing a straight-ahead lead vehicle (a radius set by width alone) or over-triggering on adjacent-lane traffic (a radius set by the vehicle’s half-diagonal); the decomposition in (4) is required for correctness, not merely convenient, a fact confirmed empirically during development (Section IV).

C. Fallback manoeuvre

On triggering, the vehicle executes the minimum-risk manoeuvre defined by UN Regulation No. 157 (Automated Lane Keeping Systems) §5.5.1 [10]: decelerate inside the current lane with a deceleration demand not exceeding 4.0 m/s^2 , lane-keeping control otherwise unchanged, holding the vehicle stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0t, 0)$ from the moment of triggering.

IV. EXPERIMENTAL SETUP

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [8] from Argoverse 2 Motion Forecasting and nuScenes. Non-ego traffic is controlled reactively (rule-based, IDM), not replayed open-loop from the log: pure log replay was found to roughly triple the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its logged trajectory — precisely the deviation a fallback manoeuvre produces. Reporting results under non-reactive replay would therefore misrepresent the cost of stopping, which is a central quantity this paper reports.

B. Harm labelling

A rollout is labelled harmful using the nuPlan closed-loop safety definition [9]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at 0.1 s resolution, not evaluated while the ego is stationary and not evaluated against tracks behind the ego. Fault attribution follows the nuPlan convention: a collision the ego suffers while stationary, or while being struck from behind, is *not* at-fault and is never counted as a missed intervention; it is instead recorded as an *induced collision* — the concrete, measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot, run online inside the closed loop so that its input reflects the actual — possibly ego-deviated — simulator history rather than the original log, which is necessary once traffic is reactive. [Predictor training data, checkpoint selection, and validation minADE to be finalised in the camera-ready version alongside Section V.]

D. Splits and pre-registration

Design decisions were made on a held-out development split, disjoint from both the calibration split used to select $\hat{\lambda}$ and the test split used only for the reported evaluation. The harm definition, risk level α , confidence level δ , and the refutation criteria for every hypothesis below were fixed and recorded before any calibration- or test-split scenario was evaluated.

V. RESULTS

Results in this section are pending the full-scale calibration and test-split evaluation ($n \approx 2,000$ scenarios per split) and are not yet reported. Diagnostic runs at small scale ($n \leq 300$) were used exclusively to validate the pipeline described in Sections III–IV — in particular, to identify and correct two errors in the trigger-score geometry of (4) before any pre-registered evaluation — and are not reported here as results.

A. *H1: does open-loop ranking predict closed-loop ranking?*
[Pending.]

B. H2: does calibrated triggering dominate hand-tuned thresholds?

[Pending.]

C. H3: does open-loop coverage collapse translate to closed-loop harm?

[Pending.]

VI. DISCUSSION

[To be completed alongside Section V. Planned content includes: sensitivity of $\hat{\lambda}$ to decision latency and decision rate; the CVaR-style tail-severity extension motivated by [7], replacing the mean missed-intervention rate in (3) with a tail bound on a continuous time-to-collision shortfall; and the limits of the reactive-IDM traffic model as a proxy for real driver response to a decelerating vehicle.]

VII. CONCLUSION

[To be completed once Section V is populated.]

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